

Frank — AMC Upper Primary **STRETCH TAIL Q26–Q30**

30 Jun 2026 | ANSWER KEY, method & the trap — for the parent

Q26. [L2] Answer: 234 [cross-number / digit-property · real 2025 Q26 (M2) + hidden condition · difficulty 5 vs real ceiling 5 · 7 marks]

What is the smallest three-digit number with all of these properties: divisible by 6; contains the digit 3; has no digit 0; has all different digits; has exactly 1 odd digit?

★ Method: Apply the strong filters first (divisibility, no-zero, distinct), then check the SUBTLE last condition before you commit — don't stop at the first multiple you find.

The trap: a solver who skips “has exactly 1 odd digit” stops at 132 — the planted decoy.

Q27. [L3] Answer: 448 [organised counting (residue classes) · scaled twin of 2013 Q27 (M5) · difficulty 21 vs real ceiling 12 · 7 marks]

In how many ways can you choose 3 different numbers from 2 to 22 so that their sum is a multiple of 3?

★ Method: Bucket all numbers by remainder mod 3; the valid triples are 'all-from-one-bucket' + the matching cross-bucket combinations.

Why it's L3: $C(21,3) = 1330$ triples — impossible to list. Sort the numbers into remainder buckets mod 3 and count the bucket combinations.

Q28. [L3] Answer: 22 [cross-number / digit-property · scaled twin of 2025 Q26 (M2) · difficulty 6 vs real ceiling 5 · 8 marks]

How many four-digit numbers have all of these properties: divisible by 12; has all different digits; has no digit 0; has exactly 2 odd digits; has a digit sum divisible by 9; contains the digit 8?

★ Method: Don't hunt for examples — partition by the binding condition (digit sum / parity) and count each class.

Why it's L3: ~9000 four-digit candidates — you cannot list them. Count structurally: fix the digit-sum/parity buckets and use divisibility.

Q29. [L3] Answer: 79 [organised counting (digit positions) · scaled twin of 2024 Q28 (M5) · difficulty 250 vs real ceiling (n/a) · 8 marks]

You write the 250 numbers 4, 8, 12, ... up to 1000. How many times do you write the digit '0'?

★ Method: Go place-column by place-column: count how often the digit appears in the units, then tens, then hundreds across the whole list.

Why it's L3: 250 terms up to 1000 — you can't write them out. Count the digit '0' one place-column at a time (units, tens, hundreds).

Q30. [L3] Answer: 9 [cross-number / digit-property · scaled twin of 2025 Q26 (M2) · difficulty 6 vs real ceiling 5 · 8 marks]

How many four-digit numbers have all of these properties: divisible by 13; has all different digits; has no digit 0; has exactly 2 odd digits; has a digit sum divisible by 9; contains the digit 6?

★ Method: Don't hunt for examples — partition by the binding condition (digit sum / parity) and count each class.

Why it's L3: ~9000 four-digit candidates — you cannot list them. Count structurally: fix the digit-sum/parity buckets and use divisibility.

Every answer is solver-verified (unique). L2 keeps real-paper difficulty but plants a decoy equal to the common slip; L3 is deliberately pushed ABOVE the real-paper ceiling and out of hand-enumeration range. Topics target Frank's tail gaps: organised enumeration (M5) and stacked-constraint cross-number (M2).